Amrit Srivastava

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Education

Michigan State University GPA: 3.7
Bachelor of Science in Computer Science, Minor in Data Science *May 2026*
Coursework: Object Oriented Design, Algorithm Engineering, Compilers, Web Development, AI/ML
Involvement: 3x Hackathon Winner, Engineer MSU Rocketry, Resident Assistant

Experience

AI Robotics Research Assistant | *Action Lab* June 2025 – Jan 2026

- Evaluated the safety and performance of robot navigation algorithms by developing a realistic city-scale environment in NVIDIA Isaac Sim with high-density vehicle and pedestrian traffic
- Developed ROS2 middleware that connected simulated robots to external computer and navigation vision algorithms, allowing the team to "plug-and-play" research algorithms without rebuilding the simulation, reducing compute time
- Implemented support for navigation and object detection with LiDAR, Radar and RGB by leveraging ROS nodes
- Authored detailed documentation with annotations and pictures to accelerate team adoption of internal tools

Data Science Intern | *iSON Xperiences Ltd.* July 2023 – Aug 2023

- Conducted an EDA and trained a machine learning model on a 1.85 million item dataset to identify potential paying customers with 87% accuracy, enabling prioritized call center outreach and increased income
- Architected and deployed a secure data pipeline with Selenium web scraping; used Python to script and automate ERP data ingestion, saving 1.5 hours/week and reducing human error

Student Library Assistant | *Digital Multimedia Center MSU Library* Jan 2023 – Present

- Automated library procurement by using Python and Selenium to source item prices, saving 2 hours fortnightly
- Enhanced the library games catalog by enriching metadata, enabling advanced sorting and improving patron experience
- Developed Rust application to sort spreadsheets by item call number, addressing limitations of Excel-based system
- Coordinated front-desk operations; facilitated room and media checkouts, maintained a 10,000+ item catalog ensuring resource accessibility and high patron satisfaction

Projects

Machine Performance Prediction | *AkzoNobel Senior Capstone*
Full-stack, Machine Learning, Databases, React, Docker

- Increased machine uptime by leveraging machine learning to predict equipment failure from real-time sensor data
- Trained neural network with Tensorflow; deployed model and database on site using Docker cutting operational cost
- Developed Next.js frontend with mobile and desktop support, leveraging caching and compression to decrease latency

Rocket Communication System | *MSU Rocketry Team*
C++, Low Latency, Linux, System Architecture, Embedded

- Developed multi-threaded C++ low-latency with XBee radio for rocket-to-mission-control telemetry exchange
- Architected a self-healing Linux environment via systemd supervision, implementing automated recovery logic and watchdog timers ensuring high availability during flight

OpenFiche | *Spartahack 2025 (Winner - Specialized Mastery)* www.reada.wiki
Python, React, Natural Language Processing, Algorithm Engineering

- Built and deployed a search engine that surfaces human-authored content by leveraging natural language processing and adapting Google's graph based PageRank search algorithm to demote low-quality autogenerated pages
- Built a lightweight, research focused web browser that displayed browsing history as an undirected graph, a popular feature that was eventually ported to Google Chrome

Technical Skills

Languages & Frameworks: Python, C++, Rust, Swift, TypeScript/JavaScript, React, SQL, Hadoop
Skills: Software Development, AI/ML, Computer Vision, Natural Language Processing, Linux Systems, Robotics